PDS Practical-5

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Aim: Data Visualization with Matplotlib: Data visualization on a dataset using matplotlib and seaborn libraries

• Scatter plot

• Line plot

• Bar plot

• Histogram

• Box plot

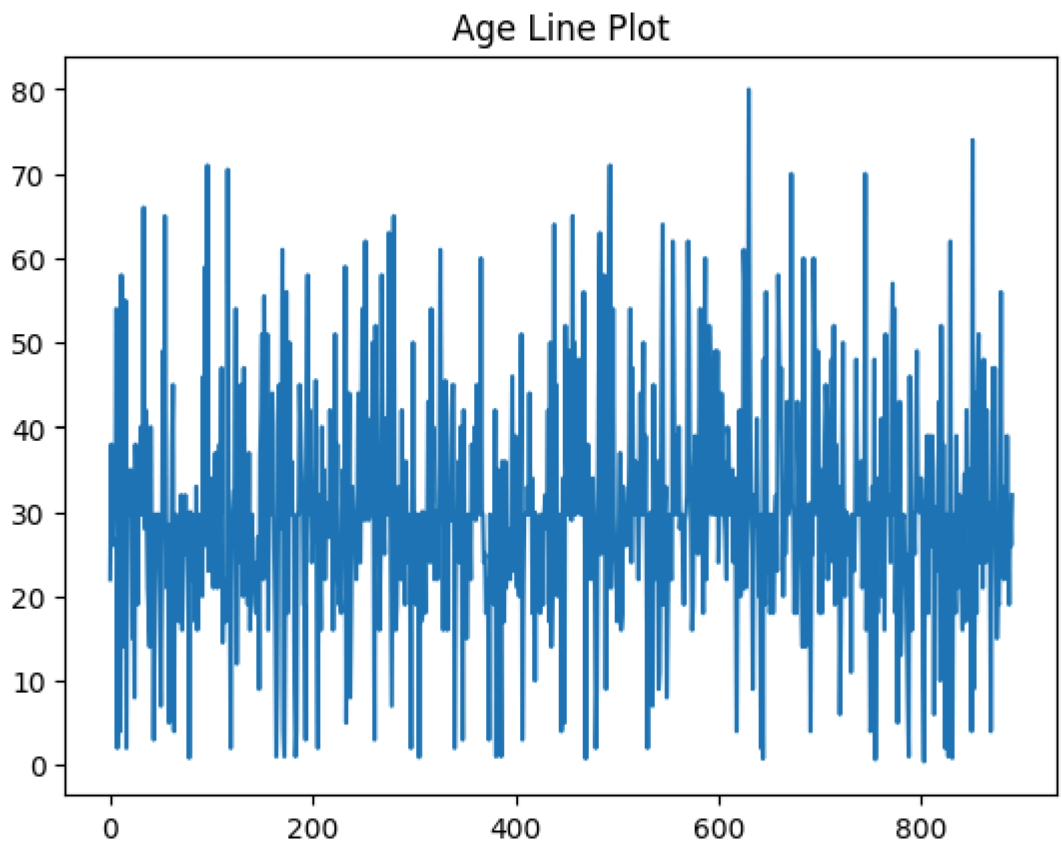
• Pair plot

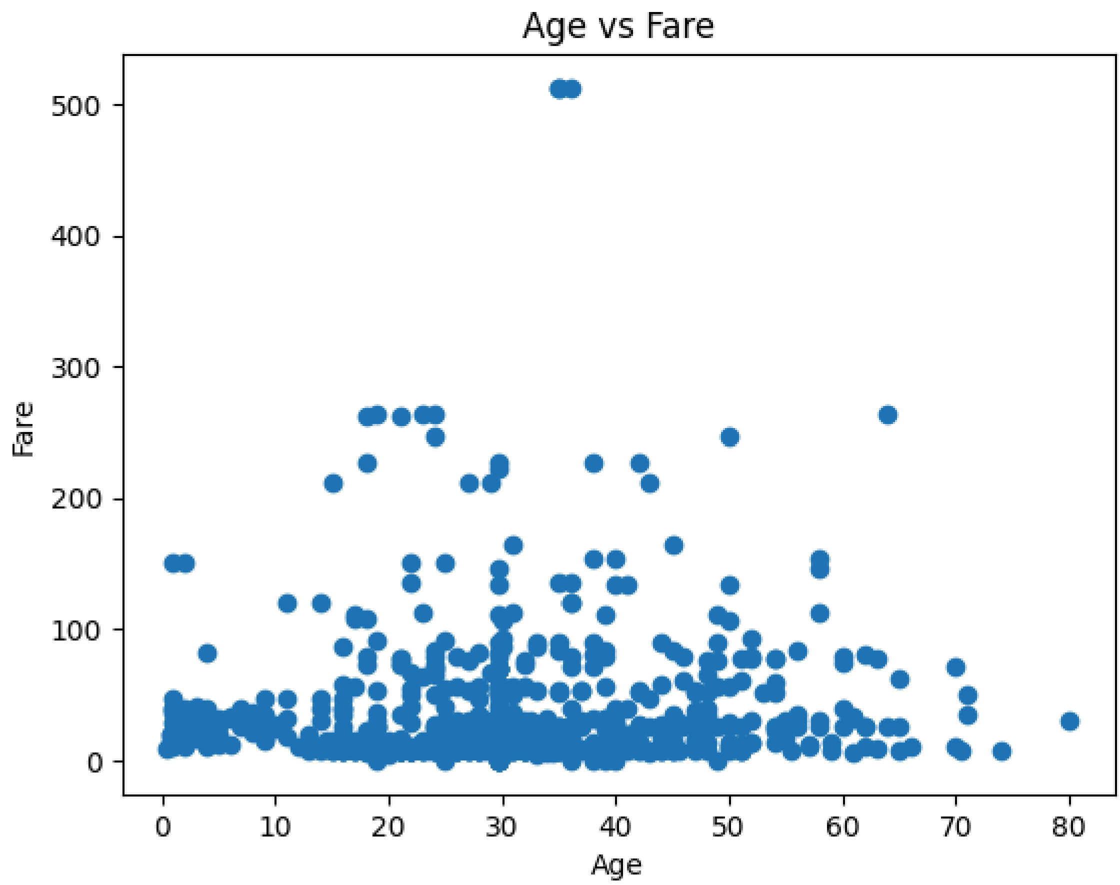
Theory

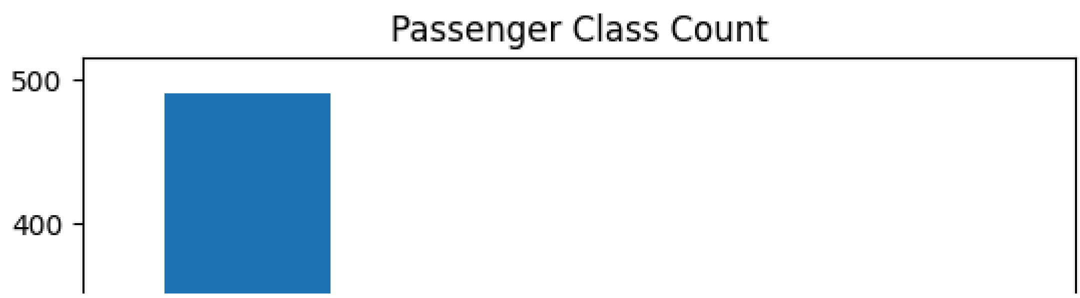
Data visualization is the graphical representation of data using plots, charts, and graphs, which helps in understanding patterns, trends, and relationships within the dataset. In this practical, Python libraries such as Matplotlib and Seaborn are used for creating various types of visualizations.

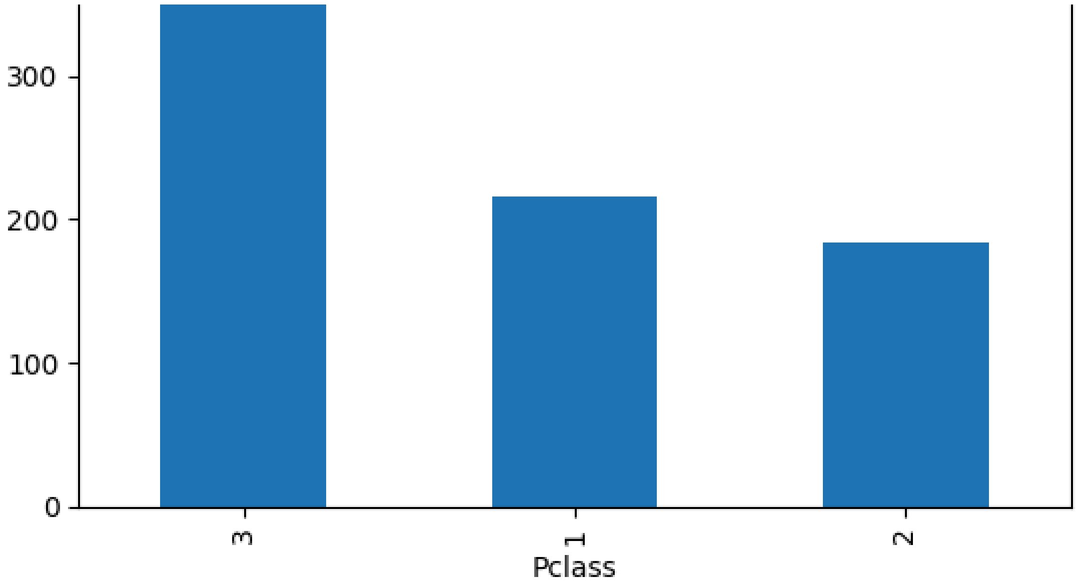
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv("https://raw.githubusercontent.com/datasciencedojo/datasets
df.head()
df["Age"] = df["Age"].fillna(df["Age"].mean())
df.drop("Cabin", axis=1, inplace=True)
plt.plot(df["Age"])
plt.title("Age Line Plot")
plt.show()
plt.scatter(df["Age"], df["Fare"])
plt.xlabel("Age")
plt.ylabel("Fare")
plt.title("Age vs Fare")
plt.show()
df["Pclass"].value_counts().plot(kind='bar')
plt.title("Passenger Class Count")
plt.show()
plt.hist(df["Age"])
```

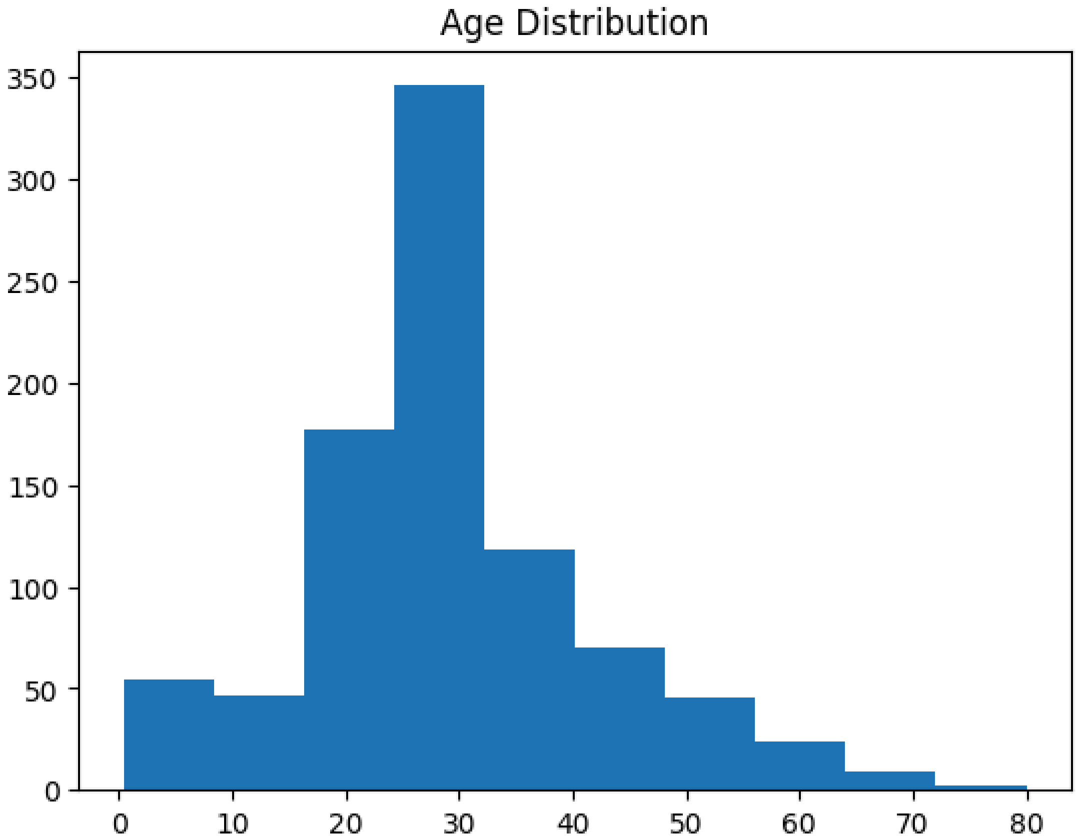
```
plt.title("Age Distribution")
plt.show()
sns.boxplot(x=df["Age"])
plt.title("Age Box Plot")
plt.show()
sns.pairplot(df)
```

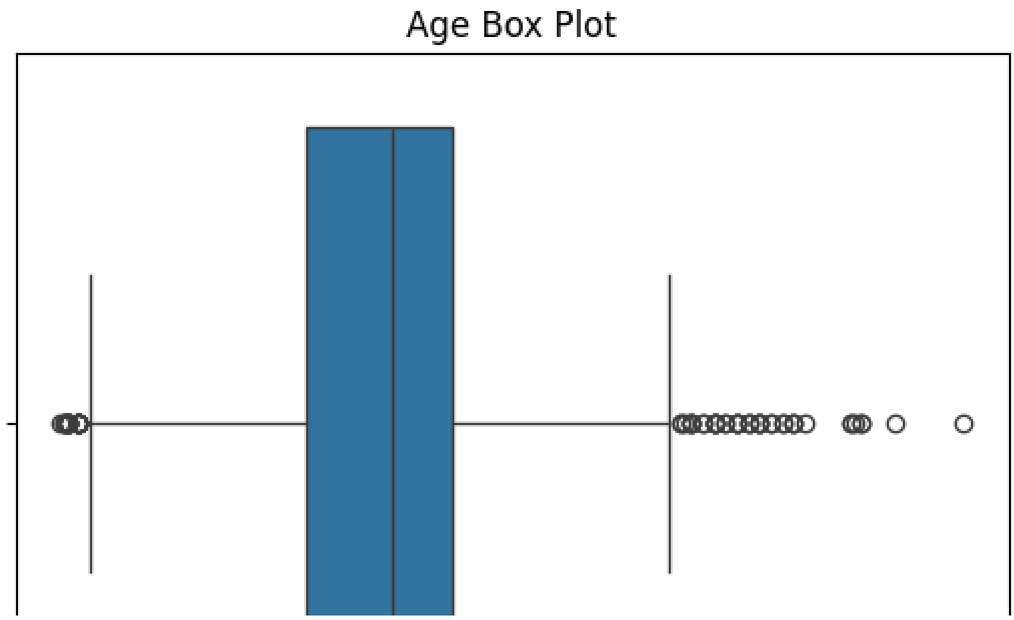

Age Line Plot
80
70
60
50
40
30
20
10
0
0
200
400
600
800

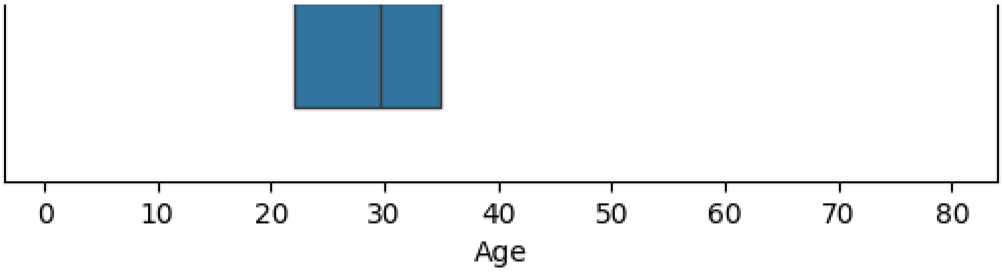

Age vs Fare
Fare
500
400
300
200
100
0
0
10
20
30
40
50
60
70
80
Age


Passenger Class Count
500
400

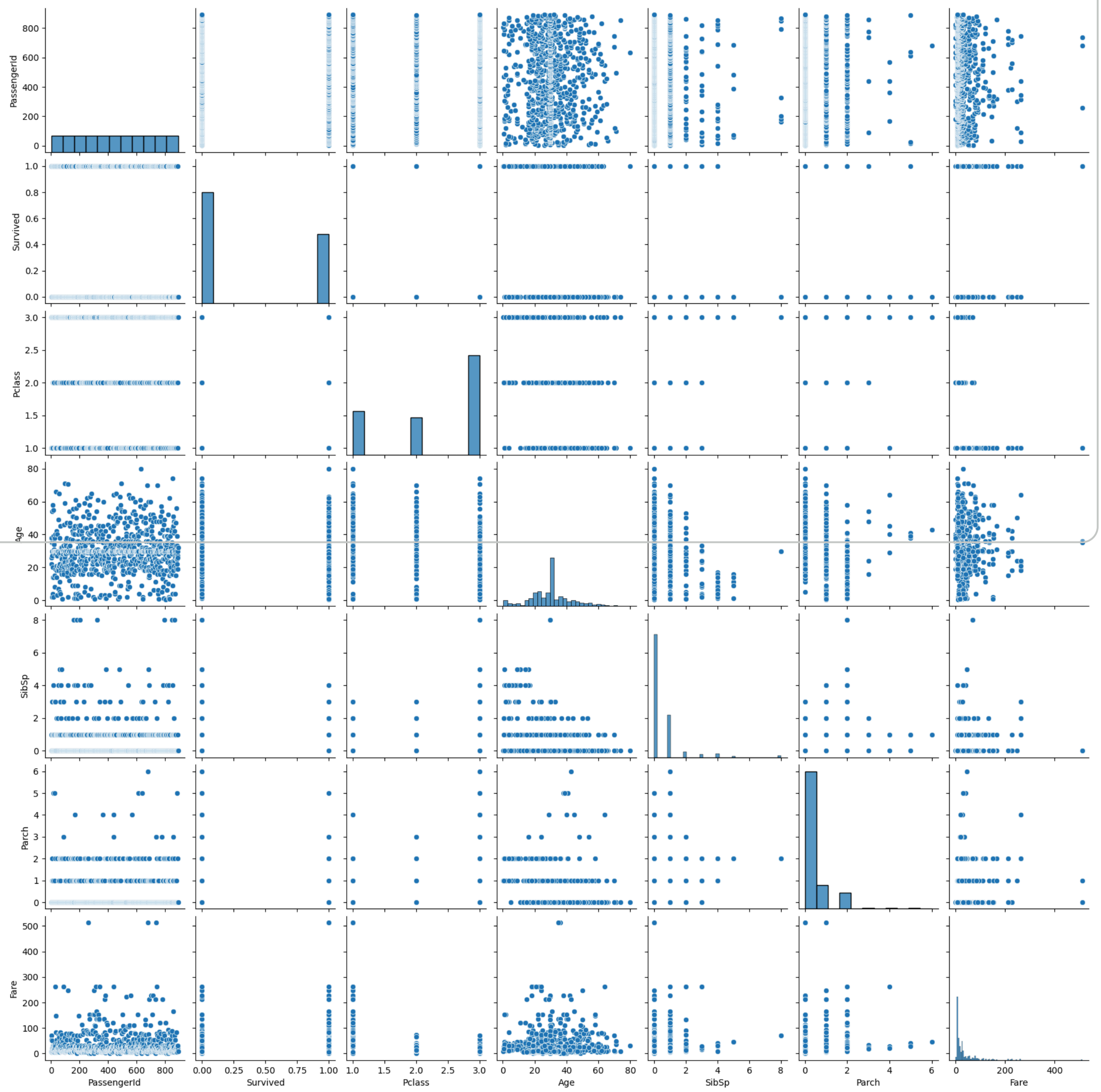

300
200
100
0
3
1
2
Pclass


Age Distribution
350
300
250
200
150
100
50
0
0
10
20
30
40
50
60
70
80


Age Box Plot



<seaborn.axisgrid.PairGrid at 0x7ea2f033c4a0>



Q1. Importance of Data Visualization & Role in EDA

Data visualization is crucial for several reasons:

1. Understanding Data at a Glance: It allows for quick comprehension of large and complex datasets by presenting information in an easily digestible visual format.
2. Identifying Patterns and Trends: Visualizations help in uncovering hidden patterns, trends, outliers, and relationships that might not be apparent in raw numerical data.
3. Facilitating Decision Making: By providing clear insights, data visualization empowers better-informed decisions across various fields.
4. Communication: It serves as an effective tool to communicate findings and insights to both technical and non-technical audiences.

Q2. Difference between Matplotlib and other libraries

Matplotlib is a foundational plotting library for Python, offering extensive control over every aspect of a plot. Here's how it generally differs from other popular visualization libraries like Seaborn, Plotly, and Bokeh:

Matplotlib:

1. Low-level control: Provides very granular control over plotting elements (figures, axes, lines, text, etc.), making it highly customizable.
2. Steep learning curve: Due to its detailed control, it can be more verbose and require more code for complex plots.
3. Static plots: Primarily designed for creating static, publication-quality plots.
4. Foundation: Many other Python plotting libraries, including Seaborn, are built on top of Matplotlib.

Seaborn:

1. High-level interface: Provides a higher-level interface for drawing attractive and informative statistical graphics.
2. Built on Matplotlib: Seaborn uses Matplotlib for its underlying plotting functionality but simplifies many common statistical plot types.

Plotly:

1. Interactive plots: Specializes in creating interactive, web-based visualizations that can be zoomed, panned, and hovered over.
2. Wide range of chart types: Supports a vast array of chart types, including 3D plots, scientific charts, and financial charts.

Bokeh:

1. Interactive and web-focused: Similar to Plotly, Bokeh focuses on creating interactive plots, dashboards, and data applications for web browsers.